Geometric Properties of the Voronoi Tessellation in Latent Semantic Manifolds of Large Language Models

Marshall Brett
MARS Labs
loganbrett27@gmail.com

Abstract

We present a comprehensive empirical study of Voronoi tessellation geometry in the representation space of Qwen3.5-4B-Base, validating and extending the latent semantic manifold framework of Mabrok (2026). Using float32 margin recomputation to eliminate bfloat16 quantization artifacts, we achieve $R^2 = 0.9997$ on the linear scaling law of the expressibility gap — the strongest confirmation of Theorem 10.5 to date. We establish definitive baseline metrics (gap coefficient $\alpha = 0.762$, entropy-margin Spearman ρ = -0.648) and demonstrate that a geometric regularization loss is correlated with cross-entropy at the final layer ($\rho = 0.836$), while exhibiting a sign reversal at layers 24-28 that reveals a mid-layer geometric ambiguity regime.

We then show that the Voronoi tessellation of a converged model is reshapable through targeted post-hoc intervention, contrary to our initial negative results. We compare two margin refinement methods across a dose-response curve: direct margin maximization and Fisher information distance maximization. Both methods find the same ~16,300 correctable positions per 256K evaluated, but differ critically in collateral damage. Margin maximization damage escalates with λ_MRP (flip ratio degrades from 2.9x to 0.4x at λ_MRP=1.0), while Fisher information damage remains constant at ~5,300 positions across the fully validated λ_MRP range (0.15-0.6). Fisher λ_MRP=0.6 achieves +28% median margin improvement with zero downstream benchmark degradation — a geometric reorganization that compresses the expressibility gap without changing benchmark means. Additional runs at Fisher λ_MRP=1.0 and λ_MRP=2.0 continue improving the margin statistics while benchmark means remain near-flat. Qwen-native frequency and token-class audits show that these gains are highly concentrated in head and structural tokens: at λ_MRP=0.6 the 100+ frequency bucket accounts for 84.0% of net corrections and the structural token class accounts for 65.9%; by λ_MRP=2.0 those shares rise to 92.1% and 75.0%. A single instruct-model follow-up shows the same concentration after fine-tuning, with lower flip ratio (1.74x) and net-negative entity-like tokens. Flat benchmarks and improved aggregate token accuracy therefore do not by themselves prove uniformly better token-level utility across the vocabulary.

Code and data: https://github.com/mellowmarshall/mrp-paper

1 Introduction

Large language models operate on discrete tokens while performing internal computations in continuous vector spaces. Mabrok (2026) formalized this continuous-discrete interface through the

latent semantic manifold framework, proving that the expressibility gap — the fraction of semantic space where the vocabulary fails to provide confident token assignments — obeys a linear volume scaling law (Theorem 10.5). This theoretical result was validated across six transformer architectures from 124M to 1.5B parameters.

We extend this validation to Qwen3.5-4B-Base (Qwen Team, 2025), a 4.2B-parameter dense transformer with Gated DeltaNet architecture and 248,320-token vocabulary — substantially larger and architecturally distinct from Mabrok's test models. We refer to these targeted post-hoc interventions collectively as margin refinement procedures (MRP): short optimization runs applied to a converged model to reshape local Voronoi geometry without full retraining. Our work makes four contributions:

1. **Methodological:** We identify and resolve a bfloat16 margin quantization artifact that systematically depresses R^2 in the log-log scaling regression, and provide a float32 recomputation technique that recovers $R^2 = 0.9997$.

2. **Analytical:** We characterize the layer-wise evolution of Voronoi geometry, discovering that geometric regularization is redundant with cross-entropy at the output layer but anti-correlated at layers 24-28 — a mid-layer geometric ambiguity regime.

3. **Empirical — Margin Maximization:** We demonstrate that direct margin maximization can reshape the Voronoi tessellation of a converged model in 200 gradient steps, achieving +53% median margin at λ_MRP=0.3 with a 2.1:1 correction-to-damage ratio. However, damage escalates super-linearly with λ_MRP, making the method destructive above λ_MRP≈0.3.

4. **Empirical — Fisher Information Distance:** We show that Fisher information distance maximization achieves comparable margin improvements (+28% at λ_MRP=0.6) with fundamentally different damage characteristics: collateral damage is constant across the fully validated λ_MRP range, downstream benchmarks are invariant through λ_MRP=0.6, and additional runs at λ_MRP=1.0 and λ_MRP=2.0 continue the monotonic margin trend while preserving near-flat benchmark means. We present a geometric argument for why this happens, while also noting that frequency and token-class audits show the gains are concentrated in head and structural tokens, with a single instruct-sample probe suggesting that entity-like regressions can emerge after fine-tuning.

2 Background

2.1 The Latent Semantic Manifold

Following Mabrok (2026), we model the contextual hidden states of a transformer at layer l as lying on a smooth k-dimensional Riemannian submanifold M^(l) of R^d, where d is the ambient hidden dimension. At the final layer, the unembedding matrix W maps hidden states to logit vectors, inducing a Voronoi tessellation of the manifold.

Definition 2.1 (Voronoi margin). For a hidden state h in M, the Voronoi margin is m(h) = l_{t*}(h) - l_{t**}(h), where t* and t** are the top two tokens by logit value.

Definition 2.2 (Expressibility gap). The normalized expressibility gap at threshold ε is $\eta(\varepsilon) = \mu(\{h \in M : m(h) < \varepsilon\})$ / vol(M). In the empirical setting, we estimate $\eta(\varepsilon)$ over the sampled hidden-state distribution induced by the evaluation corpus, so vol(M) should be read as the normalization

constant for the sampled manifold region rather than an exact geometric volume integral over the full latent manifold.

Theorem 2.3 (Mabrok, 2026; arXiv:2603.22301, Theorem 10.5). Under regularity conditions on M and the margin function, $\eta(\varepsilon) = \alpha \cdot \varepsilon + O(\varepsilon^2)$ as $\varepsilon \to 0+$, where $\alpha = H^{k-1}(\partial V) / (\lambda_geom \cdot vol(M))$. We reserve λ_geom for this manifold-side constant and use λ_MRP below for experimental loss weights.

Proof sketch. The margin function $m(h) = \ell_\{t^*\}(h) - \ell_\{t^{**}\}(h)$ is continuous with $m^{-1}(0) = \partial V$ (the Voronoi boundary). When $\|\nabla m\| \geq \lambda_geom > 0$ near ∂V, the coarea formula gives $\mu(G_\varepsilon) = \int_0^\varepsilon \int_\{m=c\} \|\nabla m\|^{-1} dH^{k-1} dc$. Since $\{m=c\}$ is $O(c)$-close to ∂V for small c, $H^{k-1}(\{m=c\}) = H^{k-1}(\partial V) + O(c)$ and $\|\nabla m\|^{-1} = \lambda_geom^{-1} + O(c)$. Substituting and integrating yields $\mu(G_\varepsilon) = (\varepsilon/\lambda_geom) \cdot H^{k-1}(\partial V) + O(\varepsilon^2)$. Dividing by vol(M) gives $\eta(\varepsilon) = \alpha \cdot \varepsilon + O(\varepsilon^2)$. The empirical test is a log-log regression of $\hat{\eta}(\varepsilon)$ on ε; slope ≈ 1 confirms linearity.

2.2 Margin Maximization

Direct margin maximization computes the loss L_margin = -E[m(h)] over positions where the margin falls below a threshold, encouraging the model to push representations away from Voronoi boundaries:

$$L = L_CE + \lambda_MRP \cdot L_margin$$

where L_margin = -mean(m(h) | m(h) < threshold). Margin maximization is computationally cheap: the only overhead beyond standard cross-entropy is a top-2 sort of the logit vector at each position. This makes it a natural baseline against which to compare more expensive geometric objectives.

Algorithm 1: Margin Maximization Loss

Input: logits $L \in \mathbb{R}^{B \times S \times V}$, threshold τ

```
top2 ← TopK(L, k=2)                    // [B, S, 2]
m ← top2[:, :, 0] − top2[:, :, 1]      // Voronoi margin
gate ← {(b, s) : m[b, s] < τ}          // low-margin positions only
L_margin ← −mean(m[gate])
return L_margin
```

2.3 Fisher Information Distance

Let $W \in R^{V \times d}$ be the unembedding matrix, where V is vocabulary size and d is hidden dimension, and let $\Sigma_p \in R^{V \times V}$ be the softmax covariance diag(p) - pp^T where p = softmax(Wh). The induced Fisher metric on hidden space is then $G(h) = W^T \Sigma_p W \in R^{d \times d}$, providing a theoretically grounded distance metric on the representation manifold. The squared Fisher distance between two token embeddings at a given hidden state is:

$$d_Fisher(w_i, w_j; h)^2 = (\tilde{w}_i - \tilde{w}_j)^T G(h) (\tilde{w}_i - \tilde{w}_j)$$

where $\tilde{w}_t = w_t / \|w_t\|$ are L2-normalized embedding rows (normalizing to unit vectors ensures the Fisher distance measures directional separation rather than norm differences; this is appropriate when embedding norms are approximately uniform, as is the case for Qwen3.5-4B where the coefficient of variation of $\|w_t\|$ across the vocabulary is <5%). The normalized embedding difference $\tilde{w}$_i -

$\tilde{w}$_j is treated as a virtual perturbation direction in hidden state space; G(h) then measures the resulting KL divergence in the output distribution to second order, so d_Fisher quantifies how distinguishable tokens i and j are to the model at position h. Note that this distance is h-dependent through the softmax covariance Σ_p. In practice, we restrict to the top-k tokens: Σ_p is replaced by the k×k submatrix Σ_k = diag(p_k) - p_k p_kˆT where p_k is the renormalized softmax over the top-k logits, and W is replaced by the corresponding k×d submatrix W_k. This top-k restriction computes the Fisher metric of the renormalized top-k distribution rather than the full softmax — an approximation that overweights pairwise distances among competitive tokens and ignores the contribution of the long tail. We expect this to be adequate in practice since the top-k tokens dominate the local geometry, but it is an approximation. Fisher MRP maximizes pairwise Fisher distances between the top-k tokens, weighted by their probabilities:

L_Fisher = -E_h[Σ_{i≠j ∈ top-k} p(t_i|h) · p(t_j|h) · d_Fisher($\tilde{w}$_i, $\tilde{w}$_j; h)]

Unlike margin maximization, which pushes along a fixed axis (the top1-top2 direction), Fisher maximization moves representations along the direction of maximum information in the model's own geometry. The cost is higher: Fisher MRP requires constructing the k×k covariance matrix, computing pairwise embedding differences, and evaluating the quadratic form at each position. In practice with k=5, this adds roughly 3–5x overhead per step compared to margin maximization, though both are dominated by the forward pass through the base model.

Algorithm 2: Fisher Information Distance Loss

Input: logits $L \in \mathbb{R}^{B \times S \times V}$, unembedding $W \in \mathbb{R}^{V \times d}$, k

```
top_k_logits, top_k_idx ← TopK(L, k)                    // [B, S, k]
p_k ← Softmax(top_k_logits)                             // renormalized top-k probs
W_k ← Normalize(W[top_k_idx])                           // L2-normalize embeddings
Σ_k ← diag(p_k) − p_k p_k^T                             // [B, S, k, k]
for each pair (i, j) where i ≠ j:
    δ ← W_k[:, :, i, :] − W_k[:, :, j, :]               // embedding difference
    proj ← δ · W_k^T                                    // project onto top-k subspace
    d_F^2[i, j] ← proj^T · Σ_k · proj                   // Fisher distance squared
d_F ← sqrt(d_F^2)
penalty ← Σ_{i≠j} p_k[i] · p_k[j] · d_F[i, j]           // probability-weighted sum
L_Fisher ← −mean(penalty)
return L_Fisher
```

3 Experimental Setup

3.1 Subject Model

Qwen3.5-4B-Base (Qwen Team, 2025): 4.21B text parameters within a 4.54B conditional generation wrapper. Dense feedforward layers (no MoE), tied input/output embeddings, 32 transformer layers, hidden dimension 2560, vocabulary size 248,320.

3.2 Corpus and Extraction

WikiText-103 validation set (Merity et al., 2017): 2,461 usable sequences, maximum length 512 tokens, yielding 256,577 token positions. Margins computed on all positions in float32.

3.3 Float32 Margin Recomputation

The model's native bfloat16 inference produces quantized logits: bfloat16 uses a 7-bit explicit mantissa (8-bit significand including the implicit leading bit), so the representable precision varies with magnitude — at logit magnitudes around 1–8 (typical for Qwen3.5-4B), the spacing between adjacent representable values ranges from approximately $2^{-7} \approx 0.0078$ to $2^{-4} = 0.0625$. Logit differences inherit this coarse grid, collapsing 256K margin values onto only 223 unique values. The root cause is not a uniform step size but the limited mantissa precision of the bfloat16 matmul in the model's `lm_head` projection. We resolve this by extracting the final hidden state (which retains sufficient precision even in bfloat16 storage), upcasting to float32, and recomputing the lm_head projection at full precision:

logits_fp32 = hidden_states[-1].float() @ lm_head_weight.float().T

This produces 238,399 unique margin values (verified: max absolute logit difference $< 2 \times 10^{-5}$). The critical point is that the hidden states themselves are not the bottleneck — they carry enough information to recover fine-grained margins; the precision loss occurs in the final matmul. A minimal reproduction script (`scripts/diagnostics/bf16_margin_repro.py`) demonstrates this artifact and the fix on a small sample. The float32 margins expose the continuous geometric structure of the manifold that bfloat16 quantization obscures. Under bfloat16 deployment, margins at the scale of the local bfloat16 step size are operationally indistinguishable, so the fine-grained gap structure at very small ε should be read as a property of the learned geometry rather than a claim about runtime decodability.

3.4 Dose-Response Protocol

Each MRP run: 200 gradient steps, AdamW optimizer, lr=1e-5, batch size 1 (no gradient accumulation), linear warmup over 5% of steps, weight decay 0.01, seed 0. All transformer body parameters plus the tied lm_head/embedding matrix are updated (full-model scope; see Section 3.6). The training corpus is the WikiText-103 validation set (2,461 sequences, 256K positions).

Supervision status. The total loss is L = L_CE + λ_MRP · L_MRP, where L_CE is the standard causal language modeling cross-entropy and L_MRP is the margin or Fisher objective. Cross-entropy is always active with coefficient 1.0. The MRP component (margin or Fisher) depends only on the model's own logit geometry and does not use ground-truth labels, but L_CE does use the next-token labels from the corpus. We note that this means the model sees next-token labels during optimization; a `--ce-weight 0` mode is now available in the codebase to enable pure-MRP runs without CE supervision, and we plan to validate the effect under that setting.

Evaluation on the optimization corpus. The margin audits and churn analyses are computed on the same corpus used for optimization. This is intentional: the purpose of MRP is to demonstrate that the Voronoi tessellation of a converged model is malleable — that underperforming positions can be directly improved by geometric intervention in a small number of steps. The claim is not that MRP generalizes margin improvements to unseen text (a standard train/test concern), but that MRP can reshape the geometry at targeted positions with bounded collateral damage. Evaluating on the optimization corpus is the correct test of that claim: it measures whether the intervention succeeded at the positions it targeted.

The generalization question that *does* matter — whether MRP damages the model's broader capabilities — is answered by the downstream benchmarks (Section 3.5), which are evaluated on entirely separate data and remain flat across the full λ_MRP range. The comparative findings

(Fisher vs margin-max damage profiles, the corrections ceiling, the constant-damage property) are properties of the loss functions and the model's weight geometry, not of the specific corpus.

The dose-response curve as implicit control. Although we do not include an explicit λ_MRP=0 (pure CE) control run, the dose-response design itself isolates the MRP contribution. Cross-entropy weight is held constant at 1.0 across all runs; only λ_MRP varies. If CE were responsible for the margin improvements, all runs would show the same effect regardless of λ_MRP. Instead, Fisher median margin scales monotonically with λ_MRP (1.174 at 0.15 $\rightarrow$ 1.317 at 0.6 $\rightarrow$ 1.630 at 2.0), and the two loss functions produce qualitatively different damage profiles at matched λ values. Both observations can only be explained by the MRP term.

The main fully preserved sweep tests:

- **Margin maximization:** λ_MRP $\in \{0.1, 0.15, 0.3, 1.0\}$
- **Fisher information:** λ_MRP $\in \{0.15, 0.3, 0.6\}$

Each run is followed by a full fp32 margin audit and per-position comparison to baseline, yielding 256,577-position churn analysis.

Two additional higher-λ Fisher runs extend the dose-response curve:

- **Fisher information, extended range:** λ_MRP $\in \{1.0, 2.0\}$

For these points, geometry audits, benchmark evaluations, and per-position audit data are available, enabling churn, frequency, and token-class analyses. These higher-λ runs do not address the domain-specific token-value question.

3.5 Downstream Evaluation

Six loglikelihood-only tasks from lm-evaluation-harness: ARC-Challenge, HellaSwag, WinoGrande, PIQA, LAMBADA (OpenAI), TruthfulQA MC1.

3.6 Parameter Update Scope

All MRP experiments use full-model training scope (`--trainable-scope text`): all transformer body parameters plus the `lm_head` projection are updated. Qwen3.5-4B uses tied input/output embeddings (`tie_word_embeddings=True`), meaning `lm_head.weight` and `model.embed_tokens.weight` share the same tensor. Updates to the lm_head therefore simultaneously modify all token input representations — the observed effects are not isolated to output-layer Voronoi boundary reshaping but include changes to the full representation pipeline.

We chose full-model scope for the initial dose-response study because it maximizes the optimizer's freedom to redistribute margin across the manifold. However, this makes it difficult to isolate whether the observed Fisher behavior arises specifically from Voronoi boundary reshaping or from broader representational changes. Ablations with frozen backbone (e.g., updating only the final norm and/or lm_head, or using low-rank adapters) would help isolate the minimal parameter subset responsible for the geometric effects and are an important direction for future work.

3.7 Gradient Treatment of TopK Selection

Both the margin loss (Algorithm 1) and Fisher loss (Algorithm 2) use hard TopK selection (`torch.topk`). The index selection is non-differentiable: gradients flow through the selected logit

values and embedding *rows* but not through the discrete choice of *which* tokens are selected. There is no straight-through estimator or soft relaxation.

For margin maximization (k=2), the gradient pushes the top-1 logit up and the top-2 logit down at positions selected by the margin threshold gate — a hard Boolean mask that is also non-differentiable.

For Fisher MRP (k=5), the gradient flows through: (1) the softmax probabilities over the top-k logits, (2) the L2-normalized embedding rows gathered at the top-k indices, and (3) the quadratic form computing pairwise Fisher distances. Embedding rows outside the top-k receive zero gradient from the Fisher loss. The top-k set is fixed per forward pass: a token ranked 6th receives no Fisher gradient regardless of its proximity to rank 5.

We expect this hard selection to be stable in practice because: (a) at k=5, the competitive token set changes infrequently between consecutive gradient steps for a well-trained model; (b) the probability weighting `p(t_i) · p(t_j)` in the Fisher loss naturally downweights tokens near the selection boundary (rank 5 carries low probability mass); and (c) the `sqrt(clamp(..., min=1e-8))` in the distance computation prevents gradient divergence at near-zero distances. Empirically, the loss curves are smooth across all runs with no instability artifacts, but we have not systematically measured top-k set stability across steps or sensitivity to k.

4 Results

4.1 Expressibility Gap Scaling

The log-log regression of $\hat{\eta}(\varepsilon)$ on the float32 margins yields:

Metric	bf16 (original)	fp32 (recomputed)	Mabrok range
β (log-log slope)	0.957	0.912	0.873-1.117
R^2	0.927	0.9997	> 0.985
α (gap coefficient)	0.655	0.762	not reported

$R^2 = 0.9997$ is the highest reported for any model on this test. The slope $\beta = 0.912$ falls within Mabrok's observed range [0.873, 1.117], confirming Theorem 10.5 on a Gated DeltaNet architecture with 248K vocabulary. The sub-unity slope is expected: Theorem 2.3 predicts $\eta(\varepsilon) = \alpha{\cdot}\varepsilon + O(\varepsilon^2)$, and the positive higher-order correction at the finite ε values used in the regression pulls the apparent log-log slope below 1. All six of Mabrok's architectures exhibited the same effect.

4.2 Margin Distribution

Metric	Qwen3.5-4B	Mabrok range (124M-1.5B)
$m_{0.05}$	0.065	0.041-0.055
$m_{0.25}$	0.401	0.228-0.297
Median	1.030	0.487-0.641
$m_{0.75}$	2.287	0.856-1.055
$m_{0.95}$	6.153	1.484-1.756
$\Pr(m < 0.5)$	0.300	0.395-0.512

Margins approximately 2x those of Mabrok's largest model (OPT-1.3B), consistent with improved tessellation quality at the 4B scale. The ambiguity floor $m_{0.05} = 0.065$ sits slightly above Mabrok's observed range (0.041–0.055), suggesting that the 4B model partially resolves some positions that smaller models treat as irreducibly ambiguous.

4.3 Layer-wise MRP-CE Correlation and Mid-Layer Geometric Ambiguity

We project each layer's hidden state through the final lm_head to compute "virtual" margins and MRP penalties at every depth. These virtual logits measure the mismatch between intermediate representation geometry and final linear decodability; they should not be interpreted as a native intermediate classifier.

Three regimes emerge:

1. **Layers 4-20 (contextual integration):** Voronoi geometry is meaningless. Margins near zero, MRP uniformly high, no correlation with final-layer CE.
2. **Layers 24-28 (mid-layer geometric ambiguity):** CE-MRP correlation is *negative* (Spearman ρ = -0.29, n = 256,577 positions, $p < 10^{-5}$). When intermediate hidden states are projected through the final lm_head, positions that are ultimately predicted correctly often remain geometrically ambiguous at these depths. Late layers then sharpen and linearize these states into the final correct decision.
3. **Layers 31-32 (prediction crystallization):** CE-MRP correlation is strongly positive (Spearman $\rho = 0.836$, n = 256,577, $p < 10^{-5}$). Voronoi geometry aligns with prediction quality. MRP is largely redundant with CE.

All correlations in this paper are Spearman rank correlations computed over the full 256,577-position evaluation set unless otherwise noted. At this sample size, all reported correlations with $|\rho| > 0.01$ are significant at $p < 10^{-5}$.

4.4 Dose-Response: Margin Statistics

Model	Median	Δ Median	Pr(m<0.5)	Δ Pr	α	R^2
Baseline	1.030	—	0.300	—	0.762	0.9997
MM λ_MRP=0.1	1.288	+25.0%	0.236	-21.3%	0.535	0.9995
MM λ_MRP=0.15	1.366	+32.6%	0.222	-26.0%	0.494	0.9995
MM λ_MRP=0.3	1.571	+52.5%	0.180	-39.9%	0.389	0.9995
MM λ_MRP=1.0	1.932	+87.5%	0.062	-79.5%	0.130	0.9881
Fish λ_MRP=0.15	1.174	+13.9%	0.271	-9.7%	0.670	0.9994
Fish λ_MRP=0.3	1.220	+18.4%	0.263	-12.2%	0.649	0.9995
Fish λ_MRP=0.6	1.317	+27.8%	0.250	-16.7%	0.612	0.9996
Fish λ_MRP=1.0	1.350	+31.1%	0.244	-18.6%	0.599	0.9995
Fish λ_MRP=2.0	1.630	+58.3%	0.214	-28.6%	0.514	0.9996

Both methods reduce the expressibility gap monotonically with λ_MRP. The power-law fit quality (R^2) is preserved at ≥0.9994 for all runs except MM λ_MRP=1.0 ($R^2 = 0.9881$), indicating the scaling law holds under margin refinement. The 1.0/2.0 Fisher rows extend the geometry, churn, per-band, benchmark, and coarse token-value analysis.

4.5 Dose-Response: Prediction Churn

Model	Churn%	W→R	R→W	Flip Ratio	Net Corrected
MM λ_MRP=0.1	20.0%	16,358	5,716	2.9x	+10,642
MM λ_MRP=0.15	20.5%	16,390	6,065	2.7x	+10,325
MM λ_MRP=0.3	22.9%	16,680	7,910	2.1x	+8,770
MM λ_MRP=1.0	50.8%	16,719	39,108	0.4x	-22,389
Fish λ_MRP=0.15	19.3%	16,195	5,317	3.0x	+10,878
Fish λ_MRP=0.3	19.3%	16,256	5,298	3.1x	+10,958
Fish λ_MRP=0.6	19.4%	16,327	5,356	3.0x	+10,971
Fish λ_MRP=1.0	18.0%	14,912	4,994	3.0x	+9,918
Fish λ_MRP=2.0	19.3%	15,350	5,842	2.6x	+9,508

Finding 1: Corrections ceiling. W→R (wrong → right) corrections plateau near ~16,300 across margin max and Fisher through λ_MRP=0.6, then soften modestly at higher Fisher strengths (14,912 at 1.0, 15,350 at 2.0). This still supports the idea of a largely fixed "geometrically accessible" correction reservoir, but it also suggests that very high Fisher strength starts reallocating some of that accessible capacity toward more concentrated head-token cleanup rather than maximizing total corrections.

Finding 2: Fisher damage is flat through λ_MRP=0.6 and still much flatter than margin max beyond it. Fisher R→W (right → wrong) damage is 5,317/5,298/5,356 across λ_MRP = 0.15/0.3/0.6 — near-constant across dose levels (range 5,298–5,356, span <0.6% of mean). At 1.0, damage remains low (4,994) and the flip ratio stays near 3.0x; by 2.0, damage rises to 5,842 and the flip ratio declines to 2.6x, still far from the catastrophic margin-max λ_MRP=1.0 regime.

Finding 3: Fisher's geometry continues improving past λ_MRP=0.6, but the practical tradeoff changes. The λ_MRP = 1.0 and 2.0 runs continue improving median margin and Pr(m<0.5), and benchmark means remain within the same narrow band as baseline. However, frequency and token-class analyses show that the higher-λ gains are increasingly concentrated in head and structural tokens rather than uniformly distributed across the vocabulary.

4.6 Runner-Up Rotation Analysis

Approximately 25% of positions retain their top-1 prediction but change their runner-up token after MRP. We call this "runner-up rotation."

Model	Rotated%	Wider Margin%	Mean Margin Δ
MM λ_MRP=0.1	25.1%	68.6%	+0.820
MM λ_MRP=0.3	26.2%	74.8%	+0.902
Fish λ_MRP=0.15	24.7%	64.8%	+0.790
Fish λ_MRP=0.3	24.8%	67.5%	+0.840
Fish λ_MRP=0.6	24.8%	72.0%	+0.922
Fish λ_MRP=1.0	23.0%	75.0%	+0.949
Fish λ_MRP=2.0	24.1%	80.0%	+1.149

Fisher λ_MRP=0.6 already matches margin max λ_MRP=0.3 on rotation margin delta (+0.922), and the 1.0/2.0 runs continue the same widening trend (+0.949, +1.149). Fisher rotates the runner-up to a different, more naturally separable token — the model finds an alternative second-place prediction that the representation space can accommodate at greater distance.

4.7 Accuracy by Margin Band

Model	m<0.5	0.5-1	1-2	2-5	5+	Overall
Baseline	21.6% (77K)	31.5% (49K)	46.6% (56K)	78.1% (53K)	99.3% (21K)	47.1%
MM λ_MRP=0.3	21.5% (46K)	27.4% (38K)	37.5% (69K)	72.1% (71K)	99.3% (33K)	50.5%
Fish λ_MRP=0.3	22.6% (68K)	31.6% (46K)	46.6% (54K)	78.3% (56K)	99.4% (33K)	51.4%
Fish λ_MRP=0.6	22.1% (64K)	30.9% (44K)	44.6% (54K)	75.9% (60K)	99.3% (35K)	51.4%
Fish λ_MRP=1.0	22.0% (63K)	30.3% (43K)	43.3% (54K)	74.1% (61K)	99.1% (35K)	51.0%
Fish λ_MRP=2.0	21.2% (55K)	28.7% (38K)	39.3% (52K)	67.2% (72K)	98.4% (40K)	50.8%

Fisher preserves per-band accuracy well through λ_MRP=0.6 while shifting positions to higher bands. At MM λ_MRP=0.3, mid-band accuracy degrades (37.5% at 1-2 band vs baseline 46.6%). Fisher λ_MRP=0.3 maintains identical accuracy at 1-2 (46.6%) with 2K fewer positions, indicating clean band transitions rather than forced displacement. The 1.0/2.0 runs show that the band shift continues, but per-band accuracy softens beyond 0.6, especially in the 1–2 and 2–5 ranges.

4.8 Margin Expansion Distribution

Model	Positions Wider	Mean Δ	Median Δ
MM λ_MRP=0.1	65.4%	+0.507	+0.197
MM λ_MRP=0.3	72.8%	+0.653	+0.408
Fish λ_MRP=0.15	60.0%	+0.448	+0.112
Fish λ_MRP=0.3	63.1%	+0.493	+0.151
Fish λ_MRP=0.6	68.2%	+0.574	+0.232

4.9 Downstream Benchmarks

Model	ARC-C	HellaSwag	WinoGrande	PIQA	LAMBADA	TQA	Mean
Baseline	.528	.751	.709	.784	.668	.354	.632
MM λ_MRP=0.15	.535	.754	.706	.785	.683	.355	.636
MM λ_MRP=0.3	.529	.754	.696	.786	.698	.353	.636
Fish λ_MRP=0.15	.534	.754	.700	.785	.670	.351	.632
Fish λ_MRP=0.3	.535	.754	.706	.784	.671	.349	.633

Model	ARC-C	HellaSwag	WinoGrande	PIQA	LAMBADA	TQA	Mean
Fish λ_MRP=0.6	.527	.751	.710	.783	.667	.354	.632
Fish λ_MRP=1.0	.529	.756	.709	.787	.677	.346	.634
Fish λ_MRP=2.0	.529	.752	.704	.785	.674	.343	.631

Maximum deviation from baseline mean across all runs is small (means remain in the 0.631–0.636 band). Task means remain broadly flat despite substantial token-level boundary reorganization. This is consistent with gains concentrated in head and structural tokens, but does not by itself identify which token classes are benefiting. The remaining question at higher λ_MRP is therefore not benchmark recovery but whether increasingly concentrated structural/head-token gains represent an acceptable tradeoff.

4.10 Qwen Frequency-Stratified Flip Audit

To test whether the frequency concentration observed at 264M scale generalizes, we bucketed the 256,577 audited positions by target-token frequency in WikiText validation and compared the baseline audit against Fisher-polished audits at λ_MRP = 0.6, 1.0, and 2.0.

Bucket	Baseline	Fish 0.6	Δ0.6	Fish 1.0	Δ1.0	Fish 2.0	Δ2.0
1	0.283	0.294	+0.011	0.287	+0.004	0.284	+0.001
2-4	0.310	0.324	+0.014	0.320	+0.011	0.318	+0.008
5-19	0.341	0.356	+0.015	0.351	+0.010	0.347	+0.006
20-99	0.391	0.409	+0.019	0.404	+0.014	0.399	+0.008
100+	0.557	0.620	+0.063	0.616	+0.059	0.617	+0.059

The gains are positive in every bucket at λ_MRP=0.6, but heavily concentrated in the highest-frequency bucket. Net corrected positions from the 100+ bucket account for 84.0% of the total Fisher gain at 0.6, 87.5% at 1.0, and 92.1% at 2.0. The concentration becomes stronger as λ_MRP increases even though benchmark means remain flat.

The top gains are dominated by structural tokens and formatting pieces: commas, periods, spaces, quotes, hyphens, "@", and "="-like separators. The most persistent negative-net regressions at higher λ_MRP are not exotic rare fragments but common functional/content tokens such as:

and, *he*, *his*, *with*, *which*, *it*, *they*, and *their*.

Fisher is not purely a punctuation optimizer, but its largest gains come from head tokens, and its higher-λ regressions cluster in ordinary medium-frequency words.

4.11 Qwen Heuristic Token-Class Audit

To go beyond raw frequency, we replayed the same per-position audits with heuristic token-class rules applied to the decoded token text. The classification hierarchy (applied in order, first match wins):

1. **Structural:** all characters are Unicode punctuation (category P*) or symbols (category S*), or the token is whitespace-only. Examples: `,`, `.`, `"`, `@`, `=`, `--`.
2. **Numeric:** matches `^[0-9][0-9,./:%+-]*$`. Examples: `123`, `2023`, `3.14`, `50%`.

3. **Function word:** pure alphabetic token whose lowercased form appears in a fixed 101-word set of articles, prepositions, pronouns, auxiliaries, and conjunctions (e.g., *the*, *of*, *is*, *they*, *which*).
4. **Entity-like:** starts with a capital letter followed by lowercase letters (`^[A-Z][A-Za-z]+$`) or is all-caps with 2+ letters (`^[A-Z]{2,}$`). Examples: *Paris*, *John*, *USA*, *NLP*.
5. **Content word:** remaining pure alphabetic tokens (nouns, verbs, adjectives, adverbs not in the function-word set).
6. **Fragment/other:** everything else (mixed alphanumeric, subword fragments with special characters).

The full classification code and word list are in `scripts/analysis/token_class_flip_audit.py`. This is a coarse taxonomy rather than a linguistic gold standard, but it distinguishes punctuation/formatting cleanup from content-bearing gains. Approximately 33 positions per run (~0.3%) fall into the fragment/other class and are omitted from the table below.

Class	Fish 0.6 Net	Share 0.6	Fish 1.0 Net	Share 1.0	Fish 2.0 Net	Share 2.0
Structural	7,231	65.9%	7,001	70.6%	7,135	75.0%
Function	1,208	11.0%	979	9.9%	982	10.3%
Content	1,708	15.6%	1,284	13.0%	890	9.4%
Entity-like	356	3.2%	240	2.4%	82	0.9%
Numeric	435	4.0%	382	3.9%	386	4.1%

Shares are computed against the full net corrected count from Table 4.5.

The token-class view sharpens the frequency result. At λ_MRP=0.6, Fisher produces positive net corrections for every coarse class, including content words and entity-like tokens, but the gains are already dominated by structural cleanup. As λ_MRP rises, structural share increases while content-word and entity-like contributions shrink. By λ_MRP=2.0, entity-like tokens are barely net positive and content words contribute less than ten percent of the total gain.

4.12 Fine-Tuned Instruct Sample

To test whether these effects survive fine-tuning, we analyzed a single fine-tuned Qwen3.5-4B instruct-style model. This is a single sample, not a matched instruct-model sweep, so it should be read as an initial probe rather than a full replication.

Metric	Instruct Baseline	Instruct Fisher	Δ / Note
Token accuracy	45.4%	48.1%	+2.63%
Flip ratio	—	1.74x	materially weaker than base-model Fisher
Head-bucket share	—	85.0%	100+ frequency bucket share of net gain
Structural class share	—	58.8%	dominant source of gains
Entity-like net	—	-219	net negative

Metric	Instruct Baseline	Instruct Fisher	Δ / Note
6-task mean*	—	0.632	no matched baseline bundle preserved

*Post-run benchmark mean over ARC-C, HellaSwag, WinoGrande, PIQA, LAMBADA, and TruthfulQA MC1.

The instruct result is directionally consistent with the base-model findings. Head-token concentration survives fine-tuning, and the practical tradeoff may be harsher: the overall flip ratio is lower, and the entity-like class is net negative even though structural and function-word cleanup remain positive. This does not prove that fine-tuning always worsens Fisher's token-value profile, but it cautions that base-model gains should not be assumed to transfer cleanly through alignment or supervised tuning.

4.13 Preliminary Targeted and Protected Variants

To explore whether Fisher can be turned from a bulk correction instrument into a more surgical editing tool, we evaluated targeted and protected variants. These runs are heterogeneous and should be read as pilot evidence rather than a matched sweep. The goal is not to claim a solved protection recipe, but to test whether targeting or protection already improves the token-value profile.

Variant	Δ Acc	Flip Ratio	Head Share of Net	Entity-like Net
Bulk Fish λ_MRP=0.6	+4.28%	3.05x	84.0%	+356
Targeted base fix03	+2.02%	1.94x	94.1%	-61
Targeted base fix06	+1.95%	1.86x	94.9%	-105
Targeted instruct fix03	+2.58%	1.68x	83.4%	-98
Targeted instruct v2 fix03	+2.63%	1.74x	85.0%	-219

The result is mixed. Targeted Fisher variants remain net positive, but on the base model they are even more head-concentrated than bulk Fisher λ_MRP=0.6 and already flip the coarse entity-like class negative. The instruct-style targeted variants show the same pattern after fine-tuning: aggregate token accuracy still rises, yet entity-like tokens remain net negative and the flip ratio is weaker than the base-model bulk Fisher result. The current targeted recipe does not yet demonstrate the desired "high-value tail preserved, head cleaned up" behavior.

We also examined 264M-scale protection pilots using mean-margin protection and Elastic Weight Consolidation (EWC; Kirkpatrick et al., 2017):

264M Pilot	Accuracy	Flip Ratio	W→R	R→W
Mean-protect, 50 steps	0.3938	2.90x	255	88
Mean-protect, 200 steps	0.4013	2.43x	336	138
Mean-protect + EWC λ=1	0.4003	2.42x	331	137
Mean-protect + EWC λ=10000	0.4013	2.43x	336	138

These pilots suggest that shorter mean-protect passes may be somewhat safer than longer ones, but they do not show a clear EWC benefit.

5 Discussion and Future Exploration

5.1 MRP Does Work

Our initial probe in bf16 reported negative results from 100-500 step MRP training. Those results do not accurately describe the effect of margin refinement in either Margin Max or Fisher formulations, so they have been omitted. The dose-response study reveals this was a formulation problem, not a fundamental limitation. With direct margin maximization or Fisher information distance as the loss, 200 steps suffices to substantially reshape the Voronoi tessellation. The key insight was moving from the cosine-weighted embedding distance proxy to direct margin or Fisher objectives.

5.2 The Corrections Ceiling

The W→R correction count is near-invariant through the main validated range: roughly ~16,300 out of 256,577 positions across margin max and Fisher up to λ_MRP=0.6. At higher Fisher strengths it softens modestly (14,912 at 1.0, 15,350 at 2.0), consistent with part of the accessible reservoir being traded away as the objective becomes more concentrated in head-token space. The ceiling interpretation still holds: only a small fraction of positions are close enough to a better Voronoi boundary that geometry alone can fix them. The remaining incorrect positions are incorrect for reasons that geometry alone cannot address — missing knowledge, not boundary proximity.

This ceiling is a fundamental property of the model's representation quality, not of the MRP method. It establishes that post-hoc margin refinement can improve accuracy by at most ~6.4 percentage points on this model/corpus pair.

5.3 Why Fisher Damage is Constant

The critical difference between margin maximization and Fisher information distance is the direction of gradient pressure:

Margin maximization pushes along the fixed axis between the current top-1 and top-2 logits. This is efficient when the two tokens are genuinely confusable (similar semantics, similar embeddings), but creates geometric distortion when they are not — forcing separation between tokens that the representation space places near each other for good reason.

Fisher information distance moves along the direction of maximum information divergence in the model's own metric tensor. When Fisher MRP increases the margin at a position, it does so by rotating the runner-up to a different token that is more naturally separable — one that the Fisher geometry identifies as the most "different" alternative. The top-1 prediction stays the same, but the second-place alternative changes to one that the representation manifold can accommodate at greater distance without distortion.

This explains the empirical observations:

1. **Constant damage:** The rotation follows the model's natural geometry, so it doesn't create distortion regardless of magnitude.
2. **Increasing margin:** Higher λ_MRP means further rotation along the same natural direction, not harder pushing along an unnatural one.
3. **Preserved per-band accuracy:** Band transitions are "clean" — positions move to higher bands because their geometry genuinely supports wider margins after runner-up rotation, not because they were forced there.

5.4 Revised Higher-λ Reading

The main sweep (Section 3.4) covers Fisher λ_MRP through 0.6. Two additional runs at λ_MRP = 1.0 and 2.0 extend the dose-response curve:

- The geometric trend continues past 0.6: median margin rises (1.317 $\rightarrow$ 1.350 $\rightarrow$ 1.630) and Pr(m<0.5) falls (0.250 $\rightarrow$ 0.244 $\rightarrow$ 0.214).
- Benchmark means remain near-flat at λ_MRP=1.0 and 2.0.
- The stronger claim that damage stays flat and token-value remains safe is only fully supported through λ_MRP=0.6.

Three conclusions follow:

1. Fisher's geometric improvements continue beyond the primary sweep range.
2. Higher λ_MRP does not collapse benchmarks or the churn profile, but it does make the token-value distribution less uniform.
3. The practical operating ceiling is determined less by aggregate damage and more by whether the added gains at high λ_MRP justify the growing concentration in structural/head-token space.

Fisher λ_MRP is therefore not a free parameter. Geometry continued to improve through λ_MRP = 2.0, but the practical cost shifted from obvious benchmark damage to a subtler redistribution of token-level utility.

5.5 Implications for the Expressibility Gap

MRP directly compresses the expressibility gap η by widening margins: Fisher λ_MRP=0.6 reduces Pr(m $<$ 0.5) from 0.300 to 0.250 (-16.7%) and the gap coefficient α from 0.762 to 0.612 (-19.7%). Critically, the power-law scaling R^2 is preserved (0.9996 vs 0.9997), indicating that MRP reshapes the gap distribution while maintaining its functional form. The gap is compressed, not broken.

This confirms that the linear scaling law of Theorem 10.5 describes a stable geometric structure that persists under margin refinement — the Voronoi tessellation is plastic (can be reshaped) but topologically invariant (retains its scaling character).

5.6 The Redundancy Resolution

Section 4.3 established that MRP loss is redundant with CE at the final layer ($\rho = 0.836$). Yet MRP demonstrably improves margins. How?

The high value correlation means that CE and MRP agree on *which* positions are problematic — but they disagree on *what to do about them.* CE optimizes the absolute logit of the correct token. MRP optimizes the gap between the top two logits, regardless of which is correct. The two losses therefore have different gradient directions in parameter space even when their magnitudes correlate across positions. MRP provides gradient signal at positions where CE is satisfied (correct prediction, reasonable confidence) but the margin is narrow — positions where MRP has independent information.

Note that value correlation across positions ($\rho = 0.836$) does not imply gradient alignment — two losses can agree on which positions are hard while prescribing different parameter updates. The dose-response results confirm that MRP's gradient carries information that CE's does not: margins

improve monotonically with λ_MRP in a regime where CE alone would have no reason to change them.

5.7 Vocabulary-Value Caveat

The Qwen-native frequency audit confirms the practical caveat first observed at 264M scale. Fisher improves aggregate token accuracy, but the gain is highly concentrated in the head of the frequency distribution. At λ_MRP=0.6, the 100+ bucket contributes 84.0% of total net corrected positions; at 1.0 and 2.0, that concentration rises to 87.5% and 92.1%.

The Qwen result is more nuanced than the 264M finding. At λ_MRP=0.6, every frequency bucket is still net positive, so Fisher is not purely a punctuation optimizer. The precise characterization is:

- Fisher improves every frequency bucket at the main operating point λ_MRP=0.6.
- The effect size is overwhelmingly concentrated in very high-frequency tokens.
- As λ_MRP increases, that concentration becomes more extreme.
- Benchmark means can remain flat even while token-value utility becomes less evenly distributed across the vocabulary.

The top token-level gains are dominated by punctuation and formatting pieces such as commas, periods, spaces, quotes, hyphens, @, and =. The dominant negative-net regressions at higher λ_MRP are common medium-frequency words: *and*, *he*, *his*, *with*, *which*, *it*, *they*, and *their*. The observed failure mode is not long-tail damage but a redistribution of token-level utility toward high-frequency structural tokens and away from medium-frequency content/function words as Fisher strength rises.

Prior work has shown that maximum-likelihood language modeling is frequency-biased and that performance on rare tokens is especially fragile under token-frequency imbalance (Diehl Martinez et al., 2024; Chung and Kim, 2025). Our result suggests that post-hoc Fisher refinement inherits that bias and, at higher λ_MRP, may amplify it in ways not visible in flat benchmark means.

This does **not** overturn the geometric result. Fisher clearly reshapes the margin distribution and improves aggregate token accuracy. But it weakens the stronger practical claim that Fisher is automatically a uniform quality-improver across the vocabulary.

The token-class analysis and instruct-sample probe sharpen this caveat. On the base model, λ_MRP=0.6 is broadly positive across every coarse token class, but structural tokens account for 65.9% of the net gain. By λ_MRP=2.0, structural share rises to 75.0%, while content words fall to 9.4% and entity-like tokens to 0.9%. In the single instruct-style sample, entity-like tokens are net negative (−219) even though overall token accuracy rises by +2.63%.

The remaining open question is whether this tradeoff persists on domain-relevant or benchmark-critical token subsets, and whether a frequency-aware or token-value-aware Fisher weighting can preserve the head gains without sacrificing medium-frequency content words or entity-like tokens.

5.8 Numerical Precision Implementation Note

A separate ablation at 264M scale found that Fisher distance computation is numerically sensitive to precision: the bf16→fp32 transition produced a real regime change in Fisher loss behavior, while fp32→fp64 was effectively noise-level. This does not affect the Qwen geometry tables (all computed in fp32), but it matters for reproducibility: weak Fisher results under low precision should not be

interpreted as evidence against the method before the Fisher computation path has been verified in fp32.

5.9 Toward More Precise Margin Refinement

The targeted and protected variants (Section 4.13) narrow the next question. Fisher clearly improves aggregate token accuracy; the open problem is whether it can be constrained so that gains fall on the tokens that matter most.

The current evidence suggests not yet. Targeted Fisher remains net positive, but on the base model it is more head-concentrated than bulk Fisher λ_MRP=0.6, and on instruct-style models the entity-like class is net negative across all targeted variants. The protection mechanisms are also insufficient: mean-margin protection does not rescue the token-value profile, and Elastic Weight Consolidation (EWC; Kirkpatrick et al., 2017) pilots at 264M scale show no meaningful improvement over matched no-EWC runs. We therefore view the current targeted/protected results as evidence that surgical geometric editing is a plausible research direction, not as evidence that it has been solved.

The practical implication remains important. Domain-tuned models with ambiguous language or high-value tails may benefit disproportionately from geometry-aware editing if the objective can be weighted by token value, frequency, or domain-specific risk. The current pilots expose the right failure mode and justify further work on token-value-aware Fisher weighting and better preservation objectives.

5.10 Training-Time Geometry Shaping and Scale

This paper addresses post-hoc polishing, but preliminary results at 264M and 1.1B scale suggest that training-time geometry shaping — for example, margin-oriented objectives during pretraining — may improve the manifold substrate on which later Fisher polishing operates by reducing redundancy and counteracting representational collapse.

Two concrete next steps follow. First, future polishing work should test whether post-training Fisher retains or destroys improved manifold structure induced during pretraining. Second, the experiments should be extended to larger models. Prior work has shown that larger language models can often be adapted in surprisingly low intrinsic dimension (Aghajanyan et al., 2021), which suggests more structured slack may be available for manifold refinement at scale. This is not yet a theorem about hidden-state manifolds, but it is a credible reason to expect that larger models may offer more room for geometry-aware refinement than the 4B regime studied here.

6 Related Work

Manifold structure in LLMs: Ansuini et al. (2019) discovered the intrinsic dimension hourglass pattern in CNNs. Valeriani et al. (2023) extended this to transformers. Mabrok (2026) provided the first theoretical framework connecting manifold geometry to vocabulary limitations.

Geometric optimization: Mano (Gu & Xie, 2026) projects momentum onto the Oblique manifold tangent space. Both Mano and Muon (Jordan, 2025) operate on the parameter manifold; our work targets the representation manifold directly.

Confidence calibration: Guo et al. (2017) showed modern neural networks are poorly calibrated.

MRP provides a geometric mechanism for improving calibration: wider margins correspond to more separable representations, enabling more reliable confidence estimates without temperature scaling.

Speculative decoding: Leviathan et al. (2023) showed that draft model confidence predicts verification success. MRP-widened margins would improve draft model confidence calibration, potentially increasing acceptance rates in speculative decoding.

Frequency bias and rare-token learning: Diehl Martinez et al. (2024) show that maximum-likelihood pre-training induces frequency bias and anisotropy, while Chung and Kim (2025) show that token-frequency imbalance can disproportionately reduce uncertainty on the most frequent words. Our frequency-stratified Fisher analysis (Section 4.10) can be read as a post-hoc analogue of this broader training-time effect.

Scale and intrinsic dimension: Aghajanyan et al. (2021) showed that large language models can often be adapted in surprisingly low intrinsic dimension. That result does not directly measure hidden-state manifold utilization, but it motivates our suggestion that larger models may leave more structured slack for geometry-aware refinement.

Hallucination detection: Gao et al. (2026) identified hallucination-associated neurons. Our mid-layer geometric ambiguity finding (Section 4.3) provides a geometric counterpart — positions where intermediate-layer ambiguity persists even when the final prediction is correct.

7 Conclusion

We validate Mabrok's latent semantic manifold framework on a 4B-parameter model with $R^2 = 0.9997$, the strongest reported confirmation of the linear scaling law. We demonstrate that the Voronoi tessellation of a converged model can be reshaped through targeted post-hoc intervention — contrary to the negative results expected from the MRP-CE redundancy analysis.

The central finding is that Fisher information distance MRP provides a low-damage mechanism for compressing the expressibility gap in aggregate metrics across the fully validated range (λ_MRP = 0.15 to 0.6). Within that range, Fisher MRP monotonically improves every geometric metric while maintaining constant collateral damage (~5,300 R→W out of 256K positions) and invariant downstream benchmarks. Additional runs at λ_MRP = 1.0 and 2.0 show continued margin improvement with near-flat benchmarks, but frequency and token-class audits reveal that the token-level gains are not uniformly distributed: they are increasingly concentrated in high-frequency structural tokens, while content and entity-like contributions shrink as λ_MRP rises. A single fine-tuned instruct-style sample shows the same concentration after fine-tuning and net-negative entity-like tokens.

The practical implication is therefore narrower than a generic two-phase recipe. Fisher MRP is best viewed as a post-training geometric polishing tool whose safest operating point in the current evidence is around λ_MRP=0.6: strong aggregate gains, flat benchmarks, and still-positive gains across all coarse token classes on the base model. Higher λ_MRP increasingly reallocates utility toward head and structural tokens, and fine-tuned models may make that tradeoff harsher. The next stage of this work is not another global-average sweep but token-value-aware refinement that can preserve the head-token gains without paying for them with entity-like or medium-frequency content regressions.

The targeted and protected pilots sharpen that recommendation. Domain-specific or ambiguity-heavy deployments may benefit from more surgical geometric manipulation, but the current evidence

does not yet show a reliable surgical Fisher recipe. Targeted Fisher remains strongly head-biased, mean-margin protection is insufficient, and EWC pilots do not improve the tradeoff. These results are promising as early evidence that token-value-aware refinement is possible, but do not constitute finished methods.

Limitations and future work. Several experiments would strengthen the empirical foundation. (i) *Parameter-scope ablation:* All experiments use full-model training scope. Because both Fisher and margin-max use the same scope, the *comparison* between them (constant vs. escalating damage) is valid — the scope is a shared confounder, not a differential one. However, ablations with frozen backbone or adapter-only updates would help isolate the minimal parameter subset responsible for the geometric effects. (ii) *Uncertainty estimates:* We report single-seed results. While the dose-response curve provides a form of internal replication — monotonic trends across 5–7 λ values would not survive high variance — formal confidence intervals across seeds would strengthen the claims. (iii) *Model diversity:* Results are on a single 4B model. Mabrok (2026) validated the underlying framework across six architectures; our contribution is the *intervention* study, which goes deep on one model rather than broad across many. Extension to additional model families is warranted. (iv) *Calibration metrics:* Since MRP widens margins, calibration changes (ECE, Brier score) are a natural test of practical impact beyond flat benchmark means. We have not yet computed these and expect to include them in a revision. (v) *Entropy penalty baseline:* We compare Fisher against margin maximization, but not against an entropy penalty (minimizing H(p) to sharpen the softmax distribution). An entropy penalty would widen margins as a side effect of sharpening, without any awareness of embedding geometry or the Fisher metric. The critical test is whether it exhibits Fisher's constant-damage property or margin-max's escalating damage — our hypothesis is the latter, since entropy sharpening pushes along the current distribution's gradient without runner-up rotation, but this remains unverified.

Finally, the polishing result should be read alongside a broader training hypothesis: training-time geometry shaping may improve the manifold on which post-hoc Fisher acts. Preliminary results at 264M and 1.1B scale motivate that idea, and prior low-intrinsic-dimension findings at larger model scales (Aghajanyan et al., 2021) suggest more room for structured refinement may exist than this paper alone can test. The most credible long-term path forward is to combine training-time geometry shaping with post-training token-value-aware polishing rather than treating bulk Fisher cleanup as the final answer.

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